

Concept-Conditional Cross-Tradition Binding in Semantic Embedding Space: A Method, Its Model-Robustness Limits, and a Pre-Registered Test of Mysticism Convergence

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Working preprint — Draft 8 (public release) **Date:** 2026-07-05 **Status:** Public working draft accompanying the formal Phase 3a pre-registration ([findings/phase3a-preregistration.md](#) , registered by the public commit and Zenodo DOI of this release). Phases 0–2 are exploratory and complete; Phase 3a is confirmatory, registered, and **not yet run**. Findings are reported as proof-of-concept and an invitation to extend, replicate, or refute. This draft incorporates revisions from an independent blind review of the release candidate; the experiment-level demands that review raised (lexical-overlap baselines, RSA noise ceilings) are recorded in §10 as open controls rather than silently deferred.

Changes from Draft 7 (private). Draft 7 added Phase 2 (multilingual originals) and framed its headline as a per-concept dissociation: AWARENESS convergence vocabulary-mediated, SUBSTRATE convergence translation-free. Draft 8 makes three major updates. **(1) A robustness correction (§6.9):** cross-model due-diligence (LaBSE vs OpenAI `text-embedding-3-large`) shows the per-concept cross-lingual dissociation is model-specific. Per-concept CCB conclusions fail to agree across embedding models in the multilingual regime, and on identical within-language data they reverse almost perfectly. The English Phase 0/1 results are unaffected (model-robust on both methods). **(2) The model-robust replacement (§6.10):** representational-similarity analysis (RSA) of within-language concept geometry, which never aligns embeddings across languages, agrees across models ($r \approx 0.78$) and shows a moderate holistic isomorphism of contemplative concept-geometry across original-language traditions. An equal-N control shows per-concept attribution is sample-size-confounded, so the isomorphism cannot currently be assigned to any single concept, including SUBSTRATE. **(3) The Phase 3a confirmatory design (§8), rebuilt on RSA and now public:** publishing it in this release *is* the pre-registration. Draft 8 also adds Phase 2b (§6.4b), the same-works-across-languages gradient in which AWARENESS and SUBSTRATE move in opposite directions under translation on a single fixed text pair. It is the cleanest single-model demonstration of the vocabulary-vs-structure mechanism, and it is explicitly scoped as a LaBSE-observed result.

Abstract

We introduce **concept-conditional cross-tradition binding (CCB)**, an embedding-based statistic that tests whether passages from different source traditions are more similar when both mention a shared structural concept than when only one does, over cross-tradition pairs with a permutation null. CCB is bias-aware (it avoids the shared-placeholder artifact of vocabulary substitution), tractable at $>10^4$ sentences, and, within English, cross-model replicable.

We stress-test it on the 65-year mysticism convergence debate (Stace, 1960; Katz, 1978; Forman, 1990; Hood, 1975). Across an English paraphrase corpus (Phase 0) and a verified whole-book English corpus (Phase 1a), five of seven pre-specified concepts (AWARENESS, RECOGNITION, WORLD, ULTIMATE, SUBSTRATE) bind

cross-tradition at $p \leq 0.0015$, replicating across two embedding models and two granularities and surviving a battery of lexical-overlap controls (binding retains 64–90% of its magnitude with the tag terms themselves deleted, §6.1b). **Phase 1b** (multi-translator, pre-registered) bounds the between-translator share of similarity variance at ~20%: translators differ, but the cross-tradition signal is not primarily a translator artifact. **Phase 2** removes English translation entirely, using original-language corpora across nine languages, and initially yielded a striking per-concept dissociation on LaBSE: AWARENESS convergence vocabulary-mediated, SUBSTRATE convergence translation-free, with a same-works control (Dhammapada × Tao Te Ching) in which translation *manufactures* AWARENESS convergence and *destroys* SUBSTRATE convergence.

We then report what a second embedding model does to this picture, as a methodological result in its own right: **in the multilingual regime, per-concept CCB conclusions do not survive a change of embedding model**. Across models the per-concept scores disagree (cross-language $r = -0.43$ over 7 concepts, not itself significant at that n) and on identical within-language data they reverse almost perfectly ($r = -0.96$), while both methods are model-robust on the English corpus (CCB $r = +0.88$, RSA $r = +0.97$). The per-concept dissociation is therefore a model-specific finding. The remaining multilingual candidate is holistic: RSA over within-language concept geometry, which never aligns across languages, agrees across models ($r = +0.78$), rejects a concept-label permutation null (mean cross-language RDM correlation $\approx +0.39$ – 0.44 , $p \approx .03$ – $.05$), and is reliably measured (split-half ceiling 0.93–0.95). Two controls added in this draft then discipline that claim sharply: the isomorphism shrinks to non-significance under an equal- N control, and it does **not** exceed an arbitrary word-set baseline matched for per-language prevalence (12th–14th percentile of the baseline distribution, §6.10b). The current multilingual evidence therefore does not establish contemplative-specific structural convergence in either per-concept or holistic form. A 40-language Bible×Quran baseline supplies reliability references and a diagnostic negative result: ULTIMATE is only middling between two traditions worshipping the *same God*, so CCB measures contextual co-location, not shared reference.

We make no claim about whether perennialism is correct. We claim (i) a method and its honestly-mapped validity domain: per-concept CCB within a language, holistic RSA across languages, with the controls that domain requires; (ii) a substantive negative finding: once model-robustness and prevalence-matched baselines are demanded, no multilingual cross-tradition convergence claim survives at current corpus power, and the concept-label permutation null standardly used for such claims is demonstrably too weak; and (iii) a **formally pre-registered confirmatory test** (Phase 3a) of genealogical relatedness, using genealogically independent Axial-Age traditions (pre-Buddhist China × pre-contact Greece), an age/contact gradient, tiered experiential/functional control concepts, RSA as the registered instrument, and the matched word-set baseline of §6.10b as the registered null. The registration is frozen in this release; the test has not been run. Code, corpora, and baselines are MIT.

1. Introduction

1.1 What the paper proposes

CCB operationalizes a sharper convergence claim than "everything converges": *specific structural axes bind specific traditions when those traditions discuss those axes*. For each pre-specified concept C and each cross-tradition passage pair, we compare the mean similarity of pairs where both mention C to pairs where only

one does; the difference is the binding of C , with a permutation-null p -value (§4.3). The construction avoids the failure modes of document-level similarity (register/vocabulary domination) and vocabulary substitution (shared-placeholder bias).

1.2 The mysticism convergence debate as test case

Perennialists (Stace, Forman) claim contemplatives from unconnected traditions report a shared structural description of reality; constructivists (Katz) claim apparent convergence is conceptual mediation and scholarly projection. The debate is textual, the sides make divergent predictions about observable text, and it has resisted qualitative resolution for 65 years. That makes it an apt CCB test case: hard, contested, and equipped with a literature that predicts both outcomes.

1.3 The confound ladder

English-only data cannot separate structural convergence in source content from convergence imposed by translators. The program climbs a ladder of confounds, each phase removing one: paraphrase (Phase 1a), between-translator variance (Phase 1b), translation itself (Phase 2), **the embedding model** (§6.9, a rung we did not originally draw and added after it broke), and finally genealogical relatedness (Phase 3a, pre-registered here).

1.4 What the paper does *not* claim

Nothing here bears on whether convergence reflects (a) shared reality, (b) shared cognition, (c) shared writing conventions, or (d) shared translator conventions. The method answers the prior question: *is there cross-tradition signal beyond what shared vocabulary, register, translators, embedding-model geometry, and (in Phase 3a) shared lineage can explain?* We do not claim perennialism settled or constructivism refuted.

1.5 Contributions

1. **CCB** with vectorized permutation testing (§4.3, Algorithm 1) and its validity domain mapped: model-robust within-language, model-fragile across languages (§6.9).
2. **Two translator-confound defenses**: Phase 1b variance decomposition (§6.2b); Phase 2 multilingual originals (§6.3–6.5).
3. **The same-works translation gradient** (§6.4b): on one fixed tradition pair, AWARENESS and SUBSTRATE move in opposite directions under translation. A single-model (LaBSE) design demonstration, pending second-model replication.
4. **The model-robustness result** (§6.9): per-concept multilingual conclusions do not survive a change of embedding model. This is a cautionary result for any multilingual embedding-based comparative claim, well beyond this project.
5. **Holistic RSA as the model-robust multilingual instrument, and its limits** (§6.10): equal-N and SNR analyses show per-concept attribution is sample-size-confounded, a split-half noise ceiling shows the instrument is reliable, and an arbitrary-word-set baseline (§6.10b) shows the current isomorphism is not concept-specific — motivating the stronger null the Phase 3a registration adopts.
6. **Reliability/normalization apparatus** (§4.4–4.6): gate-first protocol; a profile-fit reliability metric with built-in ground truth; per-language Δ -baseline; the LaBSE-vs-e5 anisotropy finding.

7. **A formally pre-registered confirmatory design** (§8) isolating genealogical relatedness, registered by this release.
8. **Open-source release** of code, corpora, and baselines.

2. Related work

Hutchinson et al. (2024) survey NLP on religious text, which is dominated by translation/parallel-corpus use, with little cross-tradition embedding-based structural comparison. The components have precedent: sentence embeddings (Reimers & Gurevych, 2019), permutation testing, and representational similarity analysis (Kriegeskorte, Mur & Bandettini, 2008); the assembly is the contribution. The §6.9 model-fragility finding connects to the embedding-stability literature (Antoniak & Mimno, 2018) and extends it to the multilingual per-concept regime. Debate landmarks: Stace (1960); Katz (1978); Forman (1990); Hood (1975, the M-Scale); Trivedi (2025); SEP Mysticism (2025). Multilingual embeddings: LaBSE (Feng et al., 2022); multilingual-e5 (Wang et al., 2024), tested and rejected for CCB (§4.1); OpenAI `text-embedding-3-large` as the second model (§6.9). Bridge thinkers (Bohm, Rovelli, Kastrup, Tononi/Koch, Hoffman, Tegmark, Bostrom, Lloyd, Wheeler) appear only in the Phase 0 corpus and are deferred pending verified-text sourcing.

3. Corpora and concepts

3.1 The seven pre-specified concepts

All analyses use seven structural concepts fixed before Phase 0 and carried unchanged through every phase. Tagging is by per-concept pattern dictionaries (§4.2); representative terms below are examples, not the full dictionaries (which ship in the repository).

concept	neutral definition	example tag terms
ULTIMATE	the tradition's absolute / ground-of-being	God, Brahman, Dao, the One, Ein Sof, Allah
SUBSTRATE	what underlies manifest phenomena, esp. via emptiness/non-being	emptiness, śūnyatā, 空/無, faná, ayin, nada, implicate order
AWARENESS	consciousness / the knowing capacity as topic	consciousness, awareness, rigpa, chit, nous, 識/心
WORLD	the status of the manifest/phenomenal world	world, phenomena, saṃsāra, the ten thousand things, māyā
SELF	the self and its nature or negation	self, soul, ego, ātman, anattā
RECOGNITION	liberation/awakening as recognizing what already is	awakening, enlightenment, mokṣa, nirvāṇa, realization
NONSEP	non-separability of observer and observed	nonduality, not-two, union, oneness, tat tvam asi

Five of the seven (AWARENESS, RECOGNITION, WORLD, ULTIMATE, SUBSTRATE) bind significantly in the English phases (§6.1); SELF and NONSEP do not, and NONSEP additionally has low prevalence in most corpora. The Bible×Quran reference corpus tracks only five concepts because RECOGNITION and NONSEP rarely fire in scripture (§4.5).

3.2 Phase 0 (English, paraphrase-heavy)

143 passages, 23 traditions, 68% investigator paraphrase; built for fast iteration and superseded for evidential purposes by Phase 1a.

3.3 Phase 1a (English, verified whole-book)

20 public-domain books, ~2.85M tokens, 14,173 sentences, 920 balanced passages, 11 traditions, a single named translator per source, license and provenance recorded in `corpus/books_manifest.json`.

3.4 Phase 1b (multi-translator, two source families)

Bhagavad Gītā and Tao Te Ching, three published translators each, for a within-source between-translator variance test. Pre-registered before analysis (public commit `d16fc8c`; Zenodo `v1.2-prereg-phase1b`). (§6.2b.)

3.5 Phase 2 multilingual corpora

- **Phase 2a — within-language cross-tradition**, nine languages: Chinese (Buddhist×Daoist), Arabic (Sufi×Falsafa), Hindi (Kabīr×Tulsīdās), Hebrew (Hasidic×rationalist), Spanish (Quietist×Carmelite), French (translated 3-tradition), Japanese (Buddhist×Confucian), Greek (Neoplatonist×Christian), English (Dhammapada×TTC). Sanskrit/Pali were screened and **excluded** (failed the gate, §4.5).
- **Phase 2b — same works across languages**: the Dhammapada × Tao Te Ching pair held fixed across classical Chinese (original), three English translators, and French; a second pair (Gītā × Dhammapada, EN/FR); a third, maximally-scalable pair (Dhammapada × Gospel of John) with verse-ID tag projection (EN/DE/VI proven).
- **Phase 2c — originals only**: classical Chinese, Arabic, Greek, Hindi, Spanish, Hebrew; translation-free by construction.
- **Phase 2d — harmonized**: re-tagged with one frozen cross-language dictionary plus corpus bulking.

3.6 Scripture reference corpus

A fixed parallel corpus (John, Genesis, Ecclesiastes, Qur'an) across up to 40 languages, used as a per-language **reliability/resolution reference**, never as a convergence target (§4.5–4.6, §6.7). English-projected verse tags hold concept assignment identical across languages so only the embedding varies. Extended with ancient-register Greek (Koine NT + Septuagint) and classical-Chinese (Wenli) editions for Phase 3a (§6.8).

3.7 Corpus selection is a method parameter

Corpus composition is an input that bounds claim strength; every claim in this paper is conditional on the corpora named above (§6.2, §8).

4. Method

4.1 Embedding

Phase 0/1 use OpenAI `text-embedding-3-large` and ONNX `all-MiniLM-L6-v2`. Phase 2's primary multilingual model is **LaBSE**. multilingual-e5 was **rejected for CCB**: it anisotropically collapses non-English text into a ~ 0.84 cosine cone (std ~ 0.02), far below the concept signal, while LaBSE keeps spread (~ 0.45 – 0.65). OpenAI `text-embedding-3-large` passes the same anisotropy screen on all six Phase 2c languages and serves as the second model for every §6.9–6.10 robustness analysis. We report anisotropy with every multilingual gate.

4.2 Concept tagging

Per-concept pattern dictionaries (case-insensitive regex for Latin scripts; normalized substring matching for Han, Arabic, Hebrew, Devanagari, and Greek): per-language dictionaries for Phase 2a, and a single frozen harmonized cross-language dictionary (`harmonized_concepts.py`) for Phase 2d and all §6.9–6.10 analyses, including the Greek SUBSTRATE fix (dropping ὅλη "matter", a known concept mis-mapping, §6.10). Tagging is a pre-specifiable hidden degree of freedom (§10).

4.3 The CCB statistic

$$\text{CCB}(C) = \overline{\text{sim}}(\text{both mention } C) - \overline{\text{sim}}(\text{exactly one mentions } C)$$

computed over cross-tradition passage pairs only. The contrast against "exactly one mentions C " (rather than "neither") controls for concept-mentioning passages being systematically more similar for reasons unrelated to C , such as length or elaboration. Significance is a permutation test at the passage-tag level: tag assignments are shuffled across passages (preserving the tagged count), the statistic is recomputed, and 2,000 permutations (5,000 for document-level tests) build the null; p is the fraction of permutations at or above the observed value, one-sided. Permuting tags rather than pairs respects the non-independence of pairs sharing a passage.

Algorithm 1: Concept-Conditional Cross-Tradition Binding

Input: passages $P = [p_1, \dots, p_n]$; each p_i has $\text{tradition}(p_i)$, $\text{text}(p_i)$
embedding model E producing unit-normalized vectors
concept C with pattern dictionary D_C
permutation count K
Output: binding score $\text{CCB}(C)$, p-value

1. Compute embeddings $v_i = E(\text{text}(p_i))$ for all i
2. Compute similarity matrix S where $S[i, j] = v_i \cdot v_j$
3. For each i , tag $t_i = 1$ if any pattern in D_C matches $\text{text}(p_i)$, else 0
4. Compute boolean mask M_{cross} where $M_{\text{cross}}[i, j] = 1$ iff
 $i < j$ and $\text{tradition}(p_i) \neq \text{tradition}(p_j)$
5. Compute both-have mask $M_{\text{both}} = M_{\text{cross}} \text{ AND } (t \text{ outer-and } t)$
6. Compute one-has mask $M_{\text{one}} = M_{\text{cross}} \text{ AND } (t \text{ outer-xor } t)$
7. $\text{binding_obs} = \text{mean}(S[M_{\text{both}}]) - \text{mean}(S[M_{\text{one}}])$
8. For k in $1..K$:
shuffle t to t' (preserving $\text{sum}(t)$); recompute masks;
 $\text{binding_perm}_k = \text{mean}(S[M'_{\text{both}}]) - \text{mean}(S[M'_{\text{one}}])$
9. $p = (\text{count}(\text{binding_perm} \geq \text{binding_obs}) + 1) / (K + 1)$

The vectorized implementation (`scripts/sentence_binding_vectorized.py`) keeps each permutation to a handful of numpy boolean and float operations, making the test tractable at $n = 14,173$ sentences. For sentence-level analysis, passages are split on punctuation and the same tagger is applied per sentence.

4.4 Within-language vs. cross-lingual CCB

Within-language CCB (a single language, so the model performs no cross-lingual alignment) is the cleaner test. **Cross-lingual** CCB (pool languages, restrict to cross-language pairs so the language baseline cancels) is weaker: a bitext-trained model's cross-language same-topic clustering partly reflects the model. §6.9 sharpens this from "weaker" to "model-dominated": per-concept cross-lingual CCB does not survive a change of embedding model.

4.5 Gate-first protocol + profile-fit reliability metric

Within-language gate: before any CCB in a language is trusted, the model must resolve concept structure within that language, operationalized as significant within-tradition binding for at least 6 of the 7 concepts on a within-language run. All nine Phase 2 languages passed (6–7/7); Sanskrit and Pali failed (2/7) and were excluded. **Profile-fit metric:** on the fixed Bible×Quran corpus, each language's concept-CCB profile (scale removed) is correlated with the 40-language consensus, giving a reliability check with a known-correct answer. The consensus tracks the five scripture-active concepts (AWARENESS > ULTIMATE > SUBSTRATE > WORLD > SELF; RECOGNITION and NONSEP rarely fire in scripture). 35/40 languages fit $r \geq 0.83$; outliers are low-resource (Burmese, Japanese, Indic), where the profile itself degrades rather than merely shrinking. The metric independently flagged Japanese, whose WORLD result had also failed in Phase 2a. Limits: it conflates embedding and translation quality; it is coarse (five concepts); it is scripture-register and high-resource-anchored.

4.6 Per-language Δ -baseline

Per-language resolution varies $\sim 4\times$ (§6.7), so absolute CCBs are not comparable across languages. The Bible×Quran CCB for a concept in a language(-pair) serves as the normalizer: $\Delta_C = \text{CCB}_C(\text{test}) - \text{CCB}_C(\text{baseline})$.

4.7 Document-level, substitution, and clustering analyses

Retained as secondary/descriptive. The vocabulary-substitution analysis in particular has a known shared-placeholder bias (documented in `methodology-notes.md`); CCB on unsubstituted text supersedes it.

4.8 Representational-similarity analysis (RSA) of concept geometry

For each language (or tradition-system), compute per-concept centroids from tagged chunks *in that language's own embedding space*, form the concept×concept cosine representational-dissimilarity matrix (RDM), and correlate RDM upper triangles across languages or systems (second-order isomorphism; Kriegeskorte et al., 2008). The embeddings are never aligned across languages; each geometry is measured internally and only the relational structure is compared. The null permutes concept labels within a system. One important scope note: the *harmonized tagging dictionary* is itself an investigator-made cross-language correspondence, so RSA removes the model's alignment step but not the dictionary's (§6.10, §10). This is the instrument that survives the §6.9 model-robustness test and is the registered Phase 3a primary (§8).

5. The translator confound and the two-defense response

A small anglophone-translator cohort produced most of the English translation tradition; convergent conventions would make a portion of any English-only cross-tradition similarity a property of translation, not source. Draft 5 named this the load-bearing limitation and prescribed two defenses; this draft reports both.

- **Phase 1b (between-translator variance).** On two source families with three translators each, cosine-similarity variance across pair types partitions into translator ($\sim 20\%$), source-content ($\sim 39\%$), and tradition ($\sim 41\%$) components, estimated from the ordered within-source/cross-source pair-type means. Translators are statistically distinguishable (a permutation test on translator labels gives $z \approx -17.9$, i.e., same-translator pairs are far more self-similar than the shuffled null) but the cross-tradition signal is **not primarily a translator artifact** (§6.2b).
- **Phase 2 (remove translation).** Original-language analysis attacks the broader objection that the scholar-tradition as a whole imposes conformity invisible to between-translator tests, by having no English translator at all. What survives translation-removal, once the second embedding model is consulted, is the holistic geometry result (§6.10); the per-concept dissociation does not (§6.9).

The confound that remains after translation is removed, shared *lineage/contact* among the surviving traditions, is the registered target of §8.

6. Results

Effect sizes are raw cosine differences; we avoid "σ-above-null" language because the permutation null is non-parametric. Exploratory phases report uncorrected p -values; see §6.11 for the multiple-comparisons policy.

6.1 Phase 0 / 1a (English) — recap

Five of seven concepts (AWARENESS, RECOGNITION, WORLD, ULTIMATE, SUBSTRATE) bind at $p \leq 0.0015$ in both corpora, replicating across two embedding models and two granularities; SELF and NONSEP do not bind. Canonical pair results: Mahāyāna×Theravāda AWARENESS 0.518; Advaita×Theravāda RECOGNITION 0.531 under the technical-only tagger. These English results are model-robust (§6.9: cross-model per-concept agreement $r = +0.88$ CCB, $+0.97$ RSA) and nothing in this draft walks them back; the lexical-overlap controls a blind review of this draft demanded were then run and substantially defend them (§6.1b).

6.1b Lexical-overlap controls (new)

Both-tagged passages share dictionary terms by construction, and embedding similarity is partly lexical, so positive CCB could in principle reduce to shared surface strings. Three controls were run on the Phase 1a passages (full write-up `findings/phase1a-lexical-controls.md`; script `scripts/phase1a_lexical_controls.py`). **Tag-term masking (decisive)**: deleting every dictionary-matched substring from tagged passages and re-embedding leaves all five binding concepts significant at 64–90% of their original magnitude — the binding is carried by surrounding context, not the tag strings. **Bag-of-words baseline**: a tf-idf model produces binding too, but with a *different* concept profile ($r = -0.23$ vs the embedding profile); RECOGNITION binds semantically but not lexically, and SELF binds lexically but not semantically (the embedding statistic correctly refuses a binding the lexical statistic would grant). **Frequency-matched random-word floor**: arbitrary same-prevalence word sets yield a generic positive CCB floor ($+0.004$ to $+0.021$), above which the real concepts sit at the 84th–100th percentile (WORLD and ULTIMATE decisive; AWARENESS, SUBSTRATE, RECOGNITION above the median but short of the 95th percentile at 50 draws). Consequence: raw CCB magnitudes include a generic lexical component and should be reported against the matched-random floor for the concept's prevalence; the binding itself is not a surface-string artifact.

6.2 Vocabulary-breadth mechanism — recap

Passage-level effect sizes deflated 3–4× between Phase 0 and 1a for broad-vocabulary concepts but **not** for technical-only SUBSTRATE ($+0.053 \rightarrow +0.054$); sentence-level deflation was only ~25%, confirming a casual-usage noise floor. A pre-specified technical-only-tagger test confirmed a two-component mechanism (noise floor + coverage asymmetry): RECOGNITION rose to $+0.110$; SUBSTRATE was unchanged (the control); ULTIMATE decreased because its technical vocabulary concentrates in nondual traditions, shifting pair coverage.

6.2b Phase 1b — between-translator variance decomposition

Pre-registered (commit `d16fc8c`, Zenodo `v1.2-prereg-phase1b`). The predicted variance ordering across pair types was confirmed (H1b.1); the translator share was 19.5%, below the pre-registered 35% threshold

(H1b.2); the translator effect is strongly detectable (H1b.3; the pre-registration stated that hypothesis's inequality in the inverted direction, an operationalization error documented in the Phase 1b findings file and repeated here for transparency). Conclusion: **Phase 1a's cross-tradition signal is not primarily a between-translator artifact.**

6.3 Phase 2a — the LaBSE per-concept picture

Within-language cross-tradition CCB, nine languages (LaBSE):

configuration	language type	AWARENESS	SUBSTRATE	ULTIMATE	WORLD
Chinese (Buddhist×Daoist)	orig, separate lineage	flat	binds	binds	binds
Hebrew (Hasidic×Rationalist)	orig, divergent register	flat	binds	binds	no
Greek (Neoplatonist×Christian)	orig, low-contrast	flat	(ὑλη artifact)	binds	binds
Arabic (Sufi×Falsafa)	orig, shared lineage	binds	binds	binds	binds
Hindi (Kabīr×Tulsīdās)	orig, shared lexicon	binds	binds	binds	binds
Spanish (Quietist×Carmelite)	orig, max overlap	binds (+.037)	binds	binds	binds
French (3-tradition)	translated	binds	flat	binds	binds
English (Dhammapada×TTC)	translated	binds	weak	mixed	binds
Japanese (Buddhist×Confucian)	rendered	binds	binds	binds	no

On LaBSE the pattern reads cleanly: AWARENESS binds exactly where awareness-lexicons overlap (lineage, school, or translation) and goes flat where they diverge; ULTIMATE binds within-language everywhere but is name-bound across languages; SUBSTRATE binds in the original emptiness-traditions. **§6.9 scopes this reading.** It is internally consistent and language-controlled, but per-concept multilingual CCB does not survive a change of embedding model, so the dissociation is reported as a LaBSE-geometry finding, not a model-independent fact.

6.4 Phase 2c — originals only (LaBSE)

Cross-language (translation-free): SUBSTRATE +0.0066 ($p=.004$), WORLD +0.0063 (.004), SELF +0.0042 (.02), **AWARENESS +0.0005 (.38, flat)**, ULTIMATE −0.004, RECOGNITION −0.0035. **Within-language:** SUBSTRATE +0.0334, ULTIMATE +0.0277, AWARENESS +0.0228, SELF +0.0230 (all $p<.0001$). On LaBSE, with translation banned, SUBSTRATE converges across languages and AWARENESS does not. Same scoping as §6.3. (The cross-language magnitudes are small in absolute terms; on the Δ -baseline scale of §4.6 they are comparable to the per-language Bible×Quran resolution floor, which is why the within-language column and the §6.10 holistic test carry the evidential weight.)

6.4b Phase 2b — the same-works translation gradient

The strongest single-model demonstration of the vocabulary mechanism: hold the tradition pair and the *texts* fixed (Dhammapada × Tao Te Ching) and vary only language/translation (LaBSE):

language (translation)	AWARENESS	SUBSTRATE	WORLD
classical Chinese (original)	-0.013 (flat)	+0.054 (p=.001)	+0.034
English — Legge	+0.014 (p=.012)	+0.028 (p=.05)	+0.011
English — Goddard	+0.012 (p=.02)	+0.013 (n.s.)	+0.017
English — Carus	-0.011 (n.s.)	+0.010 (n.s.)	+0.034
French — Julien	+0.021 (p<.0001)	-0.009 (flat)	+0.012

AWARENESS and SUBSTRATE move in **opposite directions** under translation. AWARENESS is flat in the original Chinese (the Buddhist 識/覺 and Daoist 心/神 lexicons do not converge) and is *manufactured* by translation, which renders both traditions with a shared mind/consciousness vocabulary. SUBSTRATE is strong in the original (空/無/無為 genuinely converge) and is *destroyed* by translation, which renders the two emptiness-vocabularies divergently. A second pair (Gītā × Dhammapada, EN/FR, translation-only) confirms the translation-side behavior. A third, maximally-scalable pair (Dhammapada × Gospel of John, verse-ID tag projection, 108-language Bible backbone) is proven in EN/DE/VI, adds German, and exposes translator-sensitivity even within English (AWARENESS binds with the Müller Dhammapada, flat with the Sujato). All of this is single-model (LaBSE); by the paper's own §6.9 standard it is a design demonstration awaiting second-model replication, which is the natural follow-on.

6.5 Phase 2d — operationalization-freeze (LaBSE)

Harmonized dictionary plus bulked corpora: cross-language SUBSTRATE +0.0059 ($p=.0025$), AWARENESS +0.0013 ($p=.21$). Within-language CCB proved corpus-composition-sensitive at small n (a Chinese cell flipped on bulking); the cross-language contrast is the stable one, on this model.

6.6 Cross-lingual pooled CCB

On LaBSE, all seven concepts bind weakly cross-language (NONSEP/SUBSTRATE strongest at $\sim +.010$ – $.012$; ULTIMATE weakest). §6.9 shows the pooled statistic is additionally composition-dominated (easy shared-lineage pairs carry it). We treat pooled cross-lingual CCB as unusable for inference; where §6.9 reports pooled values, they are used only diagnostically, to characterize model behavior.

6.7 Bible×Quran baseline — CCB is not a doctrinal-agreement detector

40 languages: AWARENESS/ULTIMATE/SUBSTRATE/WORLD bind at $+.045$ – $.049$ (40-language means), SELF ≈ 0 . **(a)** ULTIMATE is only middling **despite the literally-shared God**: mean $+.047$, below AWARENESS, top concept in only 10/40 languages. God-talk saturates scripture ($\approx 31,000$ both-tagged pairs vs SUBSTRATE's ~ 280) and ubiquity destroys discrimination. This is a within-language comparison (the same word "God" on both sides), so it is not a naming artifact. **CCB measures contextual co-location of deployment, not shared reference.** **(b)** A $4\times$ per-language resolution gradient (English $\sim .08 \rightarrow$ Arabic/Indic $\sim .02$) justifies the Δ -baseline and the reliability metric. (The English-scale value $\sim .08$ is the one quoted in §6.8's register results; per-language scales differ, which is the point of §4.6.)

6.8 Reliability and register (Phase 3a infrastructure)

Modern Greek profile-fit $r=0.86$; Chinese reproduces at $r=0.92$ (cross-check). Register run: SUBSTRATE binds at ~ 0.08 (English-scale Bible $\times$ Quran units) in **every** register including ancient Koine and classical Chinese (penalty Koine-modern -0.001 , classical-modern -0.013), so the ancient registers Phase 3a uses are reliable for concept-structure. Greek systematically under-binds AWARENESS across independent checks.

6.9 Instrument due-diligence: the model-robustness result (the correction)

Before freezing the Phase 3a design we asked whether its instrument would survive a change of embedding model, using OpenAI `text-embedding-3-large` (different training objective, not bitext-trained, passes the anisotropy screen on all six Phase 2c languages). On identical Phase 2c data and identical harmonized tags, cross-model agreement of per-concept conclusions:

statistic	cross-model agreement (LaBSE vs OpenAI)	n
CCB per-concept, cross-language (the Draft 7 dissociation)	$r = -0.43$	7 concepts
CCB per-concept, within-language (Phase 2c originals)	$r = -0.96$	7 concepts
RSA holistic isomorphism (§6.10)	$r = +0.78$	15 language pairs
CCB per-concept, English Phase 1 corpus	$r = +0.88$	7 concepts
RSA, English Phase 1 corpus	$r = +0.97$	tradition pairs

Interpretive cautions first: the per-concept correlations are over only seven points, so the cross-language $r = -0.43$ is not itself significant and should be read as "no agreement", not "significant anti-correlation"; the within-language $r = -0.96$ is the strong reversal, and even it warrants a leverage check across concepts (per-concept values ship in the repository). With that said, three consequences follow.

(1) In the multilingual regime, per-concept CCB conclusions do not transfer across models, and on the cleanest (within-language) data they reverse. On OpenAI, pooled cross-language CCB binds AWARENESS (+0.0131), ULTIMATE (+0.0112), and RECOGNITION (+0.0211; the maximally name-bound concept binding strongest is a tell that the model aligns spiritual *topics* aggressively), with SUBSTRATE middling (+0.0076). LaBSE's "AWARENESS=vocabulary / SUBSTRATE=structural" dissociation is a property of LaBSE's geometry. A granularity test on the specific Chinese $\leftrightarrow$ Greek pair adds a **concreteness axis**: concrete control concepts (eating/drinking/warfare) bind cross-language on both models while abstract ones (governance, SUBSTRATE) go flat, so cross-language CCB on abstract concepts fights an alignment penalty concrete concepts don't pay.

(2) The English Phase 0/1 findings are unaffected: both methods and both models agree monolingually. The fragility is specific to the multilingual regime. Why within-language non-English CCB flips across models remains only partially diagnosed: candidate mechanisms are per-language geometry differences between the models and a tagging-dictionary $\times$ model interaction (harmonized-dictionary quality varies by language), and we have not yet separated them. This diagnosis is open work, not a settled explanation.

(3) "Which concept converges" was, in retrospect, method-dependent across our own exploratory phases (English CCB → AWARENESS/RECOGNITION; LaBSE cross-language → SUBSTRATE; OpenAI pooled → everything; RSA → see §6.10). We report this method-dependence as a finding with a general moral: **any multilingual embedding-based comparative-culture claim made on a single model should be presumed model-specific until shown otherwise.**

6.10 The model-robust multilingual result: holistic RSA isomorphism

RSA over within-language concept geometry (§4.8) is robust exactly where CCB is fragile: LaBSE ↔ OpenAI agreement $r = +0.78$ on the cross-language-pair pattern, with similar magnitudes (mean cross-language RDM correlation $+0.39$ LaBSE / $+0.44$ OpenAI at full, unequal cell sizes; permutation $p \approx .03-.05$ at $K=7$ concepts over 6 languages). The result is robust to the Greek SUBSTRATE dictionary fix (dropping the ὑλη "matter" mis-mapping moves it only to $+0.392/+0.435$). Reading: **the overall relational arrangement of contemplative concepts is moderately isomorphic across original-language traditions, on both models, at marginal significance.**

Three controls and three caveats discipline the interpretation. **Equal-N control:** subsampling every concept×language cell to a common floor collapses per-concept differences. The apparent "SUBSTRATE is the relational outlier" (and equally its earlier "SUBSTRATE is special" CCB counterpart) is a **sample-size artifact**: rarer concepts have noisier centroids. At the cell sizes the current corpus supports ($n \leq 91$ per concept per language), the equal-N isomorphism does not clear the permutation null (best $p = .085$, OpenAI, $n=91$). The full-n headline and the equal-N result are the same finding at two power levels, and the abstract states both. **SNR/target-n curves:** the isomorphism rises monotonically with per-concept n on both models and has not plateaued at $n=91$; a linear extrapolation of the trend (shipped with the analysis scripts) puts reliable per-run significance at roughly **$n \approx 150-250$ chunks per concept per system**, which directly sets the Phase 3a corpus-sizing floor (§8), with the explicit caveat that an extrapolated, unplateaued curve is a floor estimate, not a measurement. **Granularity/facets:** sentence-level splitting does not help (the binding constraint is the rarest-concept × smallest-tradition cell, and shorter units degrade centroids); facet decomposition of SUBSTRATE is degenerate (facet sub-centroids at cosine $0.89-0.94$). Power must come from more *systems* and bigger per-system corpora.

Two remaining caveats: (i) the harmonized dictionary is itself a cross-language correspondence built by the investigator, so isomorphism attributable to the texts and to the dictionary are not yet separated; (ii) the permutation p over pooled language pairs does not model the non-independence of pairs sharing a language. The two controls a blind review of this draft demanded (noise ceiling, arbitrary-concept-set baseline) were then run, and §6.10b reports them: one clears the instrument, the other substantially demotes the finding.

6.10b The RSA controls: reliable instrument, non-specific signal (new)

Full write-up: `findings/phase2c-rsa-controls.md`; script `scripts/phase2c_rsa_controls.py`; seed-fixed, both models, Phase 2c originals.

Noise ceiling (clean). Split-half RDM reliability per language (100 resamples, Spearman-Brown corrected) is high everywhere: $0.88-0.97$ on both models. The resulting ceiling for the cross-language isomorphism is 0.93 (LaBSE) / 0.95 (OpenAI), so the published $+0.39/+0.44$ recovers $42-46\%$ of what the instrument could show. The isomorphism is not a noisy measurement of nothing; the instrument is reliable.

Arbitrary-word-set baseline (the demotion). Per language, 50 pseudo-concept sets were built: seven word sets each, sampled from that language's own vocabulary (excluding all real dictionary terms), each matched to the corresponding real concept's tagged-chunk count within $\pm 20\%$, and sampled *independently per language* so no cross-language correspondence exists. The cross-language isomorphism of these meaning-free pseudo-concept geometries averages **+0.49 (LaBSE) / +0.53 (OpenAI)** — and the real concept set falls at the **12th–14th percentile** of that distribution, below the baseline median on both models. Arbitrary prevalence-matched word groupings routinely produce as much or more cross-language RDM isomorphism than the real contemplative concepts do.

Reading. The weak claim survives: some cross-language geometric correlation exists, it is real relative to chance (permutation $p \approx .03-.05$) and relative to measurement noise (high ceiling), and it is model-robust. The strong claim dies, for now: *contemplative concept geometry is not detectably more isomorphic across traditions than arbitrary word sets of the same prevalence profile*. A specific mechanism is proposed and not yet isolated: prevalence rank order is similar across languages for content-free reasons, and centroid noise scales with sample size, so any prevalence-matched concept set inherits a shared noise-magnitude fingerprint that RSA partly reads. This mechanism also explains the equal-N collapse. Consequences: the concept-label permutation null is too weak for this family of claims, and the matched word-set baseline replaces it as the **registered null for Phase 3a** (§8). The natural follow-ups (a rank-shuffled prevalence variant to isolate the mechanism; an equal-N version of the baseline) are named in §10.

6.11 Multiple comparisons

Phases 0–2 are exploratory and report uncorrected p -values across many concept $\times$ language $\times$ model cells; individually marginal results there (e.g., SELF at $p=.02$ in §6.4, the RSA $p\approx.03-.05$) should be read as screening evidence, not confirmed findings. The confirmatory phase carries exactly one primary test (§8, H3a.1); everything else in it is tiered as gating QC, secondary, or exploratory and carries no α .

7. Discussion

7.1 What we have shown

Draft 7's contribution statement, "CCB dissociates structural from vocabulary-driven convergence," was itself dissociated by the second model. What survives is more precise and, we argue, more valuable: **a validity map for embedding-based cross-tradition comparison**. Per-concept CCB is a real, model-robust instrument *within a language* (the English results, the same-works gradients), now defended directly against the lexical-overlap mechanism (§6.1b masking). Across languages, per-concept scores belong to the model, not the traditions; the holistic RSA isomorphism is model-robust and reliably measured but is not yet distinguishable from prevalence-matched arbitrary word sets (§6.10b). The honest multilingual scorecard: instrument validated, convergence claim open, and the test that could settle it now registered with a properly hardened null (§8).

7.2 The mechanism picture that survives

The vocabulary mechanism itself is not retracted; it is re-scoped. The Phase 2b mirror-gradient (translation manufactures AWARENESS convergence and destroys SUBSTRATE convergence on fixed texts) and the vocabulary-breadth mechanism (§6.2) are concrete, replicable-by-design observations of how lexicon and translation shape embedding-space convergence, currently demonstrated on one model. What we no longer claim is the model-independent per-concept taxonomy ("structural SUBSTRATE / vocabulary AWARENESS") built on them.

7.3 What we have *not* shown

Not perennialism. Not a model-independent per-concept dissociation (§6.9). Not a contemplative-specific holistic isomorphism: the current cross-language signal does not exceed the matched word-set baseline (§6.10b). Not the elimination of **genealogical relatedness**: every tradition pair in Phases 0–2 shares lineage or contact, and that is the registered Phase 3a target. Not external metaphysical reality (§7.5).

7.4 The updated decomposition

(1) vocabulary/translation — *demonstrated* as a mechanism (Phase 2b gradient), per-concept attribution model-scoped; (2) register/style; (3) translator-tradition — *bounded* (Phase 1b, ~20%); (4) **embedding-model geometry — newly isolated** (§6.9), handled going forward by RSA plus mandatory two-model reporting; (5) genealogical relatedness — the largest remaining unbounded component (§8); (6) structural convergence — the holistic isomorphism residual, a *lower bound* given finite coverage.

7.5 The interpretive ceiling

Even a maximally successful Phase 3a establishes at most that the surviving convergence is **experientially/cognitively universal rather than culturally transmitted** (the Stace/Forman-vs-Katz crux). It does not establish access to external reality; shared experience could be shared *neurology*. The instrument reaches the former; the latter is beyond any text-only method.

8. Phase 3a: the pre-registered independence test (registered by this release; not run)

The full frozen protocol is `findings/phase3a-preregistration.md` ; its publication in this release **is** the act of pre-registration, timestamped by the public git commit and the Zenodo version DOI (Zenodo archives are immutable once published, which is what makes a self-hosted registration auditable). Summary:

Question. Genealogically independent Axial-Age spheres — pre-Buddhist China (Daoist/Confucian) × pre-contact Greece (Platonist/Aristotelian), original languages — with an age/contact gradient within each sphere and imported Chinese Buddhism as a known-diffusion internal reference. Structural universality predicts cross-sphere convergence *before* contact; diffusion predicts none. (We note the "genealogical independence" of Axial China and Greece is a historiographic premise with a literature of its own; the registration treats documented-contact level as a graded factor rather than an absolute, and the gradient design is the sensitivity analysis for it.)

Instrument. Holistic RSA isomorphism of within-language concept geometry (§4.8), the only instrument that survived §6.9. OpenAI `text-embedding-3-large` primary with pinned model version and released embeddings (mitigating proprietary-endpoint drift), LaBSE as the open corroborator; frozen harmonized and control dictionaries, chunk-level. **The registered null is the §6.10b prevalence-matched arbitrary-word-set baseline computed on the same systems** — Phase 2c showed it to be far stricter than the concept-label permutation, which is retained only as a floor; H3a.1 requires clearing both. **Per-concept claims are pre-declared exploratory** (the equal-N result).

Registered hypotheses. H3a.1 (primary): pooled pre-contact cross-sphere isomorphism exceeds the permutation null at $p < .05$. That null-rejection is the sole pass criterion; the accompanying magnitude prediction (+0.10 to +0.40, below the contacted-era reference of $\sim +0.44$ measured on the Phase 2c grid, a cross-corpus reference rather than a strict ceiling) is a falsifiable forecast, not part of the criterion. H3a.2 (secondary, declared underpowered): flat-vs-rising isomorphism across the contact gradient separates a structural floor from a diffusion increment. H3a.3 (gating QC): the controls-only RDM (eating/drinking/sleep/governance/warfare) must itself clear the null cross-sphere; if the instrument cannot see known human universals converge, the run stops and the result is declared uninterpretable rather than evidential. H3a.4 (exploratory): does contemplative-concept geometry pattern with the experiential anchors (shared-experience convergence) or the functional anchors (convergent evolution)? H3a.5: the imported-Buddhism diffusion reference.

Power honesty, frozen. The corpus is enlarged *after* registration under a frozen sizing rule (≥ 150 tagged chunks/concept/system; ≥ 3 school-systems per sphere; canon-slot selection, meaning texts are added by school/era slot from the standard canon and never selected by content inspection), with a pre-registered adequacy gate on within-sphere cells and a pre-declared descriptive fallback if the gate fails. A prior-exposure disclosure in the registration lists every adjacent computation ever run.

9. Methodological lessons (updated)

Carried forward from earlier phases: report CCB in relative terms (Δ , orderings, variance shares), never absolute cosines; gate-first per language (representation proxies rule out, not in); match tagging breadth across compared cells; build a fixed-content reliability reference with a known-correct answer. New and load-bearing: **single-model multilingual results are presumptively model-specific — always run a second, differently-trained embedding model before interpreting**; per-concept multilingual CCB is not rescued by pooling (composition-dominated) or by sentence-splitting (centroid degradation); RSA on within-language geometry is the alignment-free, model-robust formulation; per-concept attribution requires per-cell n the corpus must be *sized* for in advance (roughly 150–250 chunks per concept per system); power planning must target the rarest-concept $\times$ smallest-tradition cell, not the mean.

10. Limitations (updated)

1. **Genealogical relatedness** — unaddressed until Phase 3a runs (registered, not run).
2. **Lexical overlap as a driver of CCB — now bounded, not eliminated.** The three direct controls were run (§6.1b): masking and the bag-of-words divergence defend the English claims, and the random-word

control quantifies a generic lexical floor that raw CCB values include. Residuals: the controls are passage-level and single-model so far (sentence-level and second-model replication pending), and three concepts clear the matched-random floor at only the 84th–88th percentile at 50 draws.

3. **Model dependence** — now measured rather than unknown: everything multilingual and per-concept is LaBSE-scoped (§6.9); the RSA result is two-model, but a third, open, non-bitext model is desirable, and the within-language cross-model flip is not yet mechanistically diagnosed.
4. **RSA specificity** — the noise ceiling and arbitrary-word-set baseline were run (§6.10b): the instrument is reliable but the isomorphism does not exceed the matched baseline, so the holistic convergence claim is currently open, not established. Remaining: a language-pair-dependence-aware null; isolating the prevalence-fingerprint mechanism (rank-shuffled and equal-N baseline variants); separating dictionary-contributed from text-contributed isomorphism; and the corpus power the Phase 3a sizing rule specifies.
5. **Concept dictionaries as hidden DoF** — mitigated by harmonization and freezing, not eliminated; the harmonized dictionary is itself a cross-language correspondence (§4.8); no held-out human validation of tagging yet.
6. **Corpus/adversarial selection** — passage and work selection by a single investigator; adversarial selection by a constructivist-leaning scholar remains open.
7. **The Phase 2b gradient is single-model**, and modern computational sources (Bostrom, Tegmark, Kastrup, Bohm, Rovelli) remain Phase-0-only pending verified-text sourcing.
8. **Reproducibility of the proprietary embedding endpoint** — mitigated by releasing all embeddings and RDMs and by the open corroborator model, not eliminated.

11. Code and data availability

MIT at <https://github.com/davidredbird/concept-conditional-cross-tradition-binding>. This release adds: the Phase 2a per-language findings and gate/CCB scripts, Phase 2b gradient scripts and findings, the harmonized and control dictionaries (`harmonized_concepts.py` , `control_concepts.py`), the full Phase 3a instrument due-diligence suite (`phase3a_*` scripts: power analysis, second-model viability, granularity, RSA prototype/facet/equal-N/SNR/target-n, RSA-vs-CCB), the review-driven control suites (`phase1a_lexical_controls.py` , `phase2c_rsa_controls.py` , with findings `phase1a-lexical-controls.md` and `phase2c-rsa-controls.md` and results under `results/robustness/`), aggregate results JSONs, `findings/phase3a-preregistration.md` (the registration of record), and the Phase 3a design and due-diligence trail (`phase3a-design.md` , `phase3a-instrument-tests.md` , `phase3a-rsa-snr-sizing.md`). Corpus texts ship where licenses permit (public-domain and CC sources with attribution); restrictive-license source text (FLORES+, modern Bible/Quran translations, ctext/CBETA-restricted material) is excluded and documented. Phase 1b pre-registration at commit `d16fc8c` / Zenodo `v1.2-prereg-phase1b` .

Appendices

- **Appendix A** — Pre-specified candidate features and status: see `glossary.md` in the repository (the pre-registered feature seed) and §3.1 for the concept set used here.

- **Appendix B** — Per-concept robustness matrix across phases and models: shipped as aggregate JSONs under `results/` in the repository.
- **Appendix C** — Phase 1b pre-registration and the H1b.3 operationalization-direction note: `findings/phase1b-preregistration.md` and `findings/phase1b-multi-translator.md`.
- **Appendix D** — The Phase 3a pre-registration: `findings/phase3a-preregistration.md`, public as of this release.

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